The Art of Data Science

A Guide for Anyone Who Works with Data

Roger D. Peng & Elizabeth Matsui | Comprehensive Summary Notes + Data Science Reference Guide

1. Data Analysis as Art

Data analysis is hard to teach because, alongside its technical machinery, it has an artistic, creative component that resists being written down as a fixed formula. The authors compare it to songwriting: knowing music theory and song structure does not by itself produce a good song — a creative spark is needed to assemble the pieces.

- Donald Knuth's 1974 essay “Computer Programming as an Art”: science is what we understand well enough to teach a computer; everything else is art.
- Data analysts have many technical tools — linear regression, classification trees, deep learning — but assembling them to answer a question people care about is the art.
- Daryl Pregibon (1991): “statisticians have a process that they espouse but do not fully understand.”
- Goal of the book: not a fixed formula, but a general, repeatable process applicable across many kinds of analyses.

2. Epicycles of Analysis

Data analysis looks linear from the outside but is, in reality, highly iterative — a series of small “epicycles” that together trace the larger arc of the analysis.

Five core activities

- Stating and refining the question.
- Exploring the data.
- Building formal statistical models.
- Interpreting the results.
- Communicating the results.

The 3-step epicycle

- Set expectations — think about what you expect before inspecting data or running a procedure.
- Collect information — gather the data, literature, or expert input needed.
- Compare expectations to data and revise — if they match, move on; if not, fix the expectation, the data, or the question.

Worked example (asthma prevalence): a vague business question is refined into a specific, answerable one about uncontrolled asthma; row-count/age-range checks catch a missing data file; an unexpected gender association turns out to be a coding error, not a real finding; and the final report spawns a new analysis once the boss asks follow-up questions — showing the epicycle applies even to communication.

- A “sharp hypothesis” (e.g., “the meal will cost $30”) teaches more when checked against data than a vague one (“between $0 and $1,000”).

3. Stating and Refining the Question

Careful thought about the question is the single highest-leverage step in a data analysis — most fatal pitfalls trace back to a poorly framed question.

Six types of questions (Peng & Leek, Science)

- Descriptive — summarizes a characteristic of the data; result is a fact, not an interpretation.
- Exploratory — hypothesis-generating; looks for patterns or relationships.
- Inferential — tests whether an exploratory pattern holds in a representative sample of the population.
- Predictive — identifies what predicts an outcome, without needing to know why.
- Causal — asks whether changing one factor changes another, on average, in a population.
- Mechanistic — asks exactly how one factor produces a change in another.

Five characteristics of a good question

- Of interest to your specific audience.
- Not already answered — check the literature and internal records first.
- Grounded in a plausible framework (pepperoni near pizza sauce: plausible; pepperoni vs. yogurt: not, absent prior reason).
- Answerable with realistically available data.
- Specific — every possible answer has one clear, meaningful interpretation.

Translating a question into a data problem

- Measurement mismatch — a proxy variable that doesn't cleanly capture the real concept yields multiple plausible interpretations.
- Confounding — a third factor related to both exposure and outcome can create or hide an association.
- Bias — recall bias (differential memory of exposures) and selection bias (non-random recruitment) both distort results irreparably after the fact.

Case study: Fit on Fleek

"Joe" iterates a vague marketing objective into a specific, answerable, predictive question: "Which demographic and health characteristics identify users most likely to have chronic drowsiness (at least one episode every other day)?" — refining first for plausibility (daytime drowsiness, not sleep duration, is the better marker) and then for specificity (via colleague feedback), before recognizing it is a prediction question.

4. Exploratory Data Analysis (EDA)

EDA examines a dataset's structure, distributions, and relationships — mainly via visualization — to find problems, confirm the data can answer your question, and sketch a first-pass answer.

Nine-step checklist (2014 U.S. hourly-ozone case study)

- 1. Formulate your question — "Do eastern U.S. counties have higher ozone than western counties?"
- 2. Read in your data; watch for parsing warnings or wrong column types.
- 3. Check the packaging — confirm row/column counts and variable classes match documentation.
- 4. Look at the top and bottom (head/tail) — catches formatting glitches or stray trailer rows.
- 5. Always check your "n"s — counting catches problems; an odd timestamp traced to one off-schedule monitor, and 52 "states" turned out to include D.C. and Puerto Rico (harmless).
- 6. Validate against an external source — comparing to the 0.075 ppm national air-quality standard confirmed units and range were sensible.
- 7. Make a plot — boxplots by state flagged Puerto Rico, Georgia, and Hawaii as having unusual high-end values.
- 8. Try the easy solution first — a simple longitude cutoff (−100°) showed the west had higher mean/median ozone; the pattern survived removing the three unusual states.
- 9. Follow-up questions — Right data? Need other data (more years)? Right question (violations of the standard may be more actionable)?

5. Using Models to Explore Your Data

A statistical model provides (a) data reduction — compressing observations into a few interpretable numbers — and (b) structure, an expectation of how data were generated, including built-in randomness.

Price-survey example

- 20 responses reduced to mean ($17.20) and SD ($10.39) under a Normal model; the model can then answer new questions (e.g., ~11% would pay over $30) without re-consulting raw data.
- Checking fit: overlaying the Normal curve on the histogram revealed a poor fit (a $10 spike the curve misses; Normal allows nonsensical negative prices).
- Refining: a Gamma distribution (positive-only) fit better and changed the practical answer to ~7% paying $30+, showing model choice affects downstream decisions.

Ozone vs. temperature (linear relationships)

- A simple linear-regression expectation was compared to real 1999 NYC data and found biased — under-predicting at high temperatures, over-predicting at moderate ones.
- A loess smoother revealed the true relationship was flat until ~70°F, then rose sharply, then leveled near 90°F — a nonlinear pattern the linear model missed.

When to stop iterating

- Are you out of data? Some questions (why nonlinear?) require new or independent data to answer/replicate.
- Enough evidence to decide? The bar depends on the purpose (business decision vs. published paper).
- Does it make sense in context? Suspiciously perfect results often signal a mistake; results that clash with prior work need more scrutiny either way.
- Out of time or money? A legitimate, common, and honest stopping criterion.

6. Inference: A Primer

Inference means making a statement — with quantified uncertainty — about a population you haven't fully observed, using a sample you have.

Three ingredients

- Identify the population — the most important task; if you can't define it, you cannot infer anything about it.
- Describe the sampling process — how data moved from population to dataset, and whether that process biases representation.
- Describe a model for the population — typically parameters plus a random error term. George Box: "all models are wrong, but some are useful."

What degrades inference quality

- Selection bias — sampling systematically over/under-represents parts of the population.
- Violated independence — units influencing each other shrinks the effective sample size.
- Sampling variability — pure chance causes repeated samples to differ even under perfect methodology.

Case study: Apple Music (2015)

MusicWatch surveyed 5,000 of ~11 million free-trial users and reported 48% had stopped using the service; Apple stated the true figure was 21%. Candidate explanations: sampling variation (unlikely to explain a gap this large), selection bias in recruitment, differing definitions of "still using," and possible non-independence among respondents.

Populations come in many forms

- Time series — the "population" might be all possible years, or all stocks in the market.
- Natural processes in space (e.g., earthquakes) — modeled as a random process generating your observed data.
- Data-as-population — e.g., auditing all employee salaries: no sampling, so no inference is needed.

7. Formal Modeling

Formal modeling develops a precise mathematical specification of the question and a rigorous framework for challenging your own results.

Primary vs. secondary models

- Primary model — current best answer from exploratory work; a starting point, not a final answer.
- Secondary models — built to challenge the primary model (sensitivity analysis): add confounders, change functional form, drop outliers.

Associational analyses — three variable roles

- Outcome — the feature expected to change with the key predictor.
- Key predictor — the main factor of interest.
- Potential confounders — related to both predictor and outcome; must be identified and adjusted for.

Ad-campaign case study: a primary model estimates a $44.75/day sales lift; secondary models adding quadratic/4th-order time trends estimate $39.86 and $49.10 — evaluated on effect size, plausibility, and parsimony (prefer the simplest model telling an equally good story).

Prediction analyses

- Goal shifts to accuracy; every predictor is fair game — no notion of "confounder."
- Data split into training/test sets for honest performance evaluation.
- Credit-risk case study: 70% accuracy but only ~2.7% specificity — a single accuracy number can hide serious weaknesses; the right sensitivity/specificity trade-off depends on real-world error costs.
- Tuning parameters and more/better data are often bigger performance levers than swapping algorithms.

8. Inference vs. Prediction: Modeling Strategy

Confusing an inferential goal with a predictive goal leads to the wrong modeling approach and potentially the wrong conclusion.

- Inferential goal — estimate an association for a predictor of interest, carefully adjusting for confounders; sensitivity analyses test robustness across confounder sets.
- Predictive goal — best possible accuracy using any available predictors, with little concern for mechanism.

Case study: PM10 and mortality in NYC

- A naive model (PM10 alone) showed no association — essentially zero.
- Adding "season" revealed a real positive association — a classic Simpson's Paradox, since season relates to mortality and PM10 in opposite directions.
- Adding temperature, dew point, and NO2 kept PM10 significant, though the effect size shrank somewhat with more confounders.
- A random-forest prediction model ranked PM10 near the bottom in importance — not because there's no true effect, but because its contribution to predictive accuracy is small next to temperature or NO2.
- Lesson: a prediction-only approach might wrongly dismiss PM10; the properly adjusted inferential approach reveals a small but meaningful association — which matters given how many people are exposed.

9. Interpreting Your Results

Interpretation happens throughout an analysis, but deserves deliberate attention once modeling is complete and before communicating.

Four guiding principles

- Revisit your original question — prevents drift, reminds you of the question type, and prompts a fresh bias check.
- Focus on directionality, magnitude, and uncertainty of the primary model — not a binary significant/not-significant verdict.
- Build an overall interpretation from the totality of your analysis plus outside subject-matter knowledge.
- Consider implications — what action the result justifies, often shaped by the requesting organization's mission.

Case study: soda consumption and BMI

- Bias check: imagine the sample result is wrong and ask what recruitment/measurement quirk could explain the gap (e.g., a fitness-magazine survey, or omitting sodas like Mountain Dew from a checklist).
- Directionality: positive relationship (more soda, higher BMI), matching exploratory expectations.
- Magnitude: a raw slope of 0.28 kg/m² per ounce becomes 3.36 kg/m² per 12-oz can/day — always convert back to the units of the original question.
- Uncertainty: a 95% CI (0.15–0.42 kg/m²) is more informative than a bare p-value, since it conveys a plausible range rather than a binary label.
- Secondary models: adding income can shrink the effect (partial explanation) and/or widen the CI (often due to missing data reducing sample size) — these are distinct phenomena to check separately.
- Small effects can matter hugely at population scale (parallel: air pollution and cardiovascular events).
- Implications can justify action (e.g., a public-health campaign) even from associational, non-causal evidence, depending on organizational mission.

10. Communication

Communication is both a working tool used throughout an analysis (to gather feedback) and the final deliverable.

Three types of routine communication

- Focused fact-finding — a short, specific question; minimal background, targeted audience.
- Working through puzzling results — a small meeting spanning data-analysis and content expertise; moderate context needed.
- General feedback — sought at a natural pause (e.g., after EDA); benefits from a wide range of stakeholders and a full project walk-through.

Four concepts for effective communication

- Audience — choose people suited to the feedback needed.
- Content — tailor detail strictly to the objective (e.g., ~1 overview slide, 1–2 context slides, 5–8 substantive slides, 10–15 minutes).
- Style — informal, jargon-free; invite questions during, not just after, a presentation.
- Attitude — open, collaborative, non-defensive; a defensive tone discourages honest feedback.

11. Concluding Thoughts

The epicycle framework — set expectations, collect information, compare, revise — applies uniformly to every stage: question, exploration, modeling, interpretation, communication.

The ABCs of data analysis

- Always Be Checking — your data, assumptions, intermediate results.
- Always Be Challenging — your primary model, your interpretation, your conclusions.
- Always Be Communicating — throughout the process, not just at the end.

Mastery comes from repeated practice — with enough real-world analyses, the epicycle process becomes second nature and largely subconscious.

12. About the Authors

- Roger D. Peng — Associate Professor of Biostatistics, Johns Hopkins Bloomberg School of Public Health; co-founder of the Johns Hopkins Data Science Specialization and the Simply Statistics blog.
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Appendix: General Data Science Reference Guide

This appendix adds general, frequently asked data science concepts and definitions — useful as broader background knowledge alongside the book's specific framework.

What is Data Science?

- Data science is the interdisciplinary field that uses statistics, programming, and domain knowledge to extract insights and build predictive systems from data.
- Data Science vs. AI vs. Machine Learning: AI is the broad goal of making machines act intelligently; Machine Learning (ML) is a subset of AI where systems learn patterns from data; Data Science is the broader practice of collecting, cleaning, analyzing, and interpreting data, which often uses ML as one tool among many.
- Common roles: Data Analyst (reporting/descriptive insights), Data Scientist (modeling/inference/prediction), Data Engineer (building data pipelines/infrastructure), ML Engineer (deploying models to production).

The Data Science Lifecycle

- 1. Problem/question definition — mirrors this book's “stating and refining the question.”
- 2. Data collection — gathering data from databases, APIs, sensors, surveys, or web scraping.
- 3. Data cleaning/wrangling — handling missing values, duplicates, outliers, inconsistent formats.
- 4. Exploratory Data Analysis (EDA) — summarizing and visualizing data to find patterns.
- 5. Feature engineering — creating or transforming variables to improve model performance.
- 6. Modeling — applying statistical or machine learning algorithms.
- 7. Evaluation — measuring model performance with appropriate metrics.
- 8. Deployment — putting the model into production for real use.
- 9. Monitoring — tracking model performance over time and retraining as needed.

Types of Data

- Structured data — organized in rows/columns (e.g., spreadsheets, SQL tables).
- Unstructured data — text, images, audio, video without a fixed schema.
- Semi-structured data — JSON, XML: has some organizational tags but no strict schema.
- Quantitative (numeric) vs. qualitative (categorical) data; categorical can be nominal (no order) or ordinal (ordered).
- Time-series data — observations indexed by time, often with trend and seasonality.

Core Statistics Concepts

- Mean, median, mode — measures of central tendency; median is more robust to outliers than the mean.
- Variance and standard deviation — measures of spread/dispersion around the mean.
- Correlation — measures linear association between two variables (range −1 to 1); correlation does not imply causation.
- Probability distributions — Normal (bell curve), Binomial (successes in fixed trials), Poisson (counts of rare events), Uniform (equal likelihood).
- Hypothesis testing — null hypothesis (H0, no effect) vs. alternative hypothesis (Ha, an effect exists); a p-value estimates the probability of observing the data (or more extreme) if H0 were true.
- Confidence interval — a range likely to contain the true population parameter, with a stated confidence level (commonly 95%).
- Type I error (false positive) — rejecting a true null hypothesis; Type II error (false negative) — failing to reject a false null hypothesis.
- Central Limit Theorem — the sampling distribution of the mean approaches Normal as sample size grows, regardless of the population's original distribution.

Data Cleaning & Preprocessing

- Handling missing data — deletion, mean/median/mode imputation, or model-based imputation.
- Outlier detection — z-scores, IQR (interquartile range) method, visual inspection via boxplots.
- Feature scaling — normalization (0–1 range) or standardization (zero mean, unit variance); important for distance-based algorithms like KNN and gradient-descent-based models.
- Encoding categorical variables — one-hot encoding, label encoding, target encoding.

- Handling class imbalance — oversampling (e.g., SMOTE), undersampling, or class-weighted loss functions.

Machine Learning Basics

- Supervised learning — learns from labeled data to predict an outcome (regression for continuous outcomes, classification for categorical outcomes).
- Unsupervised learning — finds structure in unlabeled data (clustering, e.g., k-means; dimensionality reduction, e.g., PCA).
- Reinforcement learning — an agent learns by interacting with an environment and receiving rewards/penalties.
- Common algorithms — linear/logistic regression, decision trees, random forests, gradient boosting (XGBoost, LightGBM), support vector machines, k-nearest neighbors, naive Bayes, neural networks.
- Overfitting vs. underfitting — overfitting means the model fits training data (including noise) too closely and generalizes poorly; underfitting means the model is too simple to capture the underlying pattern.
- Bias-variance tradeoff — more complex models tend to have lower bias but higher variance, and vice versa; the goal is to minimize total error on unseen data.
- Cross-validation — splitting data into multiple train/validation folds (e.g., k-fold) to get a more reliable estimate of model performance.
- Regularization — techniques like L1 (Lasso) and L2 (Ridge) penalties that discourage overly complex models and reduce overfitting.

Model Evaluation Metrics

- Regression — Mean Absolute Error (MAE), Mean Squared Error (MSE), Root MSE (RMSE), R^2 (proportion of variance explained).
- Classification — Accuracy, Precision, Recall (Sensitivity), F1-score (harmonic mean of precision and recall), Specificity, ROC curve and AUC (area under the curve).
- Confusion matrix — a table of true positives, true negatives, false positives, and false negatives used to derive most classification metrics.
- Choosing a metric depends on the cost of different error types — e.g., in medical screening, high recall (sensitivity) is often prioritized over precision.

Common Tools & Languages

- Programming languages — Python (most widely used for ML/data science), R (strong for statistics), SQL (querying relational databases).
- Python libraries — pandas (data manipulation), NumPy (numerical computing), matplotlib/seaborn (visualization), scikit-learn (classical ML), TensorFlow/PyTorch (deep learning).
- Data visualization tools — Tableau, Power BI, matplotlib, ggplot2 (R).
- Big data tools — Apache Spark, Hadoop, for processing datasets too large for a single machine.
- Version control — Git/GitHub for tracking code changes and collaboration.

Deep Learning & Advanced Topics

- Neural networks — layers of interconnected nodes (neurons) that learn representations from data; deep learning uses many layers.
- CNNs (Convolutional Neural Networks) — specialized for image data, using convolutional filters to detect spatial patterns.
- RNNs/LSTMs — designed for sequential data (text, time series) by maintaining a memory of previous inputs.
- Transformers — the architecture behind modern large language models (LLMs); rely on “attention” mechanisms rather than sequential recurrence.
- Natural Language Processing (NLP) — techniques for working with text data, including tokenization, embeddings, sentiment analysis, and language models.

Frequently Asked Concepts (Quick Reference)

- Data Science vs. Statistics — statistics is a mathematical foundation; data science additionally emphasizes programming, data engineering, and applying methods at scale to real-world messy data.

- What makes a good dataset? — relevant to the question, sufficiently large, representative of the target population, and reasonably clean/complete.
- Causation vs. correlation — two variables can move together without one causing the other (confounding, reverse causation, or coincidence); establishing causation typically requires randomized experiments or careful causal-inference methods.
- Why is EDA important? — it catches data-quality problems early, builds intuition, and sets expectations before formal modeling — directly echoing this book's core framework.
- What is a p-value, really? — the probability of observing a result as extreme as (or more extreme than) what was found, assuming the null hypothesis is true; it is not the probability that the null hypothesis is true.
- Why split data into train/test sets? — to get an honest, unbiased estimate of how a model will perform on new, unseen data.
- What is feature engineering? — the process of creating new input variables (or transforming existing ones) to help a model better capture the underlying relationship with the outcome.
- What is a RAG (Retrieval-Augmented Generation) pipeline? — a system that retrieves relevant text/document chunks from a knowledge base and feeds them to a language model to ground its generated answer in specific, up-to-date, or domain-specific content.
- What is data wrangling/munging? — the broad process of cleaning, restructuring, and enriching raw data into a usable format before analysis, including joining tables, reshaping (wide vs. long), and parsing dates/strings.
- Batch vs. streaming data processing — batch processes large chunks of data at scheduled intervals; streaming processes data continuously, in near real time, as it arrives (e.g., Apache Kafka).
- What is A/B testing? — a controlled experiment comparing two (or more) versions of something (a webpage, feature, or treatment) to determine which performs better, using randomization to support causal conclusions.
- What is multicollinearity? — when two or more predictor variables in a regression model are highly correlated with each other, making it hard to isolate each variable's individual effect and inflating coefficient uncertainty.
- What is dimensionality reduction? — techniques (e.g., PCA, t-SNE, UMAP) that reduce the number of variables in a dataset while preserving as much meaningful structure/variance as possible, often used for visualization or to speed up modeling.
- What is an ensemble method? — combining multiple models (e.g., bagging, boosting, stacking) to produce a single prediction that is typically more accurate and robust than any individual model.
- What is data leakage? — when information from outside the training dataset (often from the future, or from the target itself) inadvertently informs the model, producing unrealistically good performance that fails to hold up in production.
- Parametric vs. non-parametric models — parametric models (e.g., linear regression) assume a fixed functional form with a finite set of parameters; non-parametric models (e.g., decision trees, KNN) make fewer assumptions and can grow in complexity with the data.
- What is the curse of dimensionality? — as the number of features grows, the data becomes increasingly sparse in that high-dimensional space, making distance-based methods and generalization harder without more data.
- Why do we standardize/normalize before certain algorithms? — algorithms based on distance (KNN, K-means) or gradient descent (neural networks, logistic regression) are sensitive to the scale of input features, so unscaled features with large ranges can dominate the result.
- What is a data pipeline? — an automated sequence of steps (extract, transform, load — ETL) that moves raw data from source systems into a form ready for analysis or modeling, often run on a schedule.
- What is model drift? — the gradual decline in a deployed model's performance as real-world data patterns shift away from what the model was originally trained on, requiring monitoring and periodic retraining.
- Descriptive vs. predictive vs. prescriptive analytics — descriptive summarizes what happened, predictive forecasts what is likely to happen, and prescriptive recommends what action to take in response.
- What is survivorship bias? — drawing conclusions only from the subset of data that “survived” some selection process, while ignoring the (often more informative) cases that did not — leading to overly optimistic conclusions.